Testing Halo Models for Constraining Astrophysical Feedback with Multi-Probe Modeling

arXiv: 2507.13317

Is your model ready for Stage-IV surveys?

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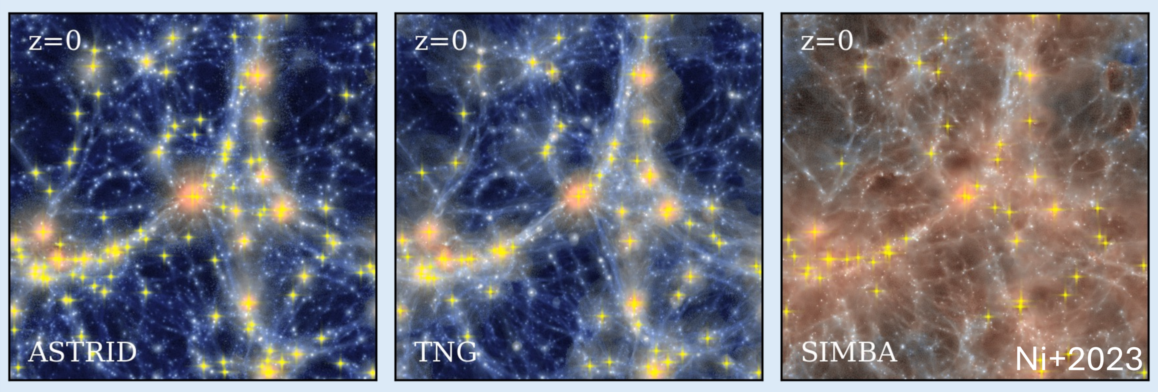
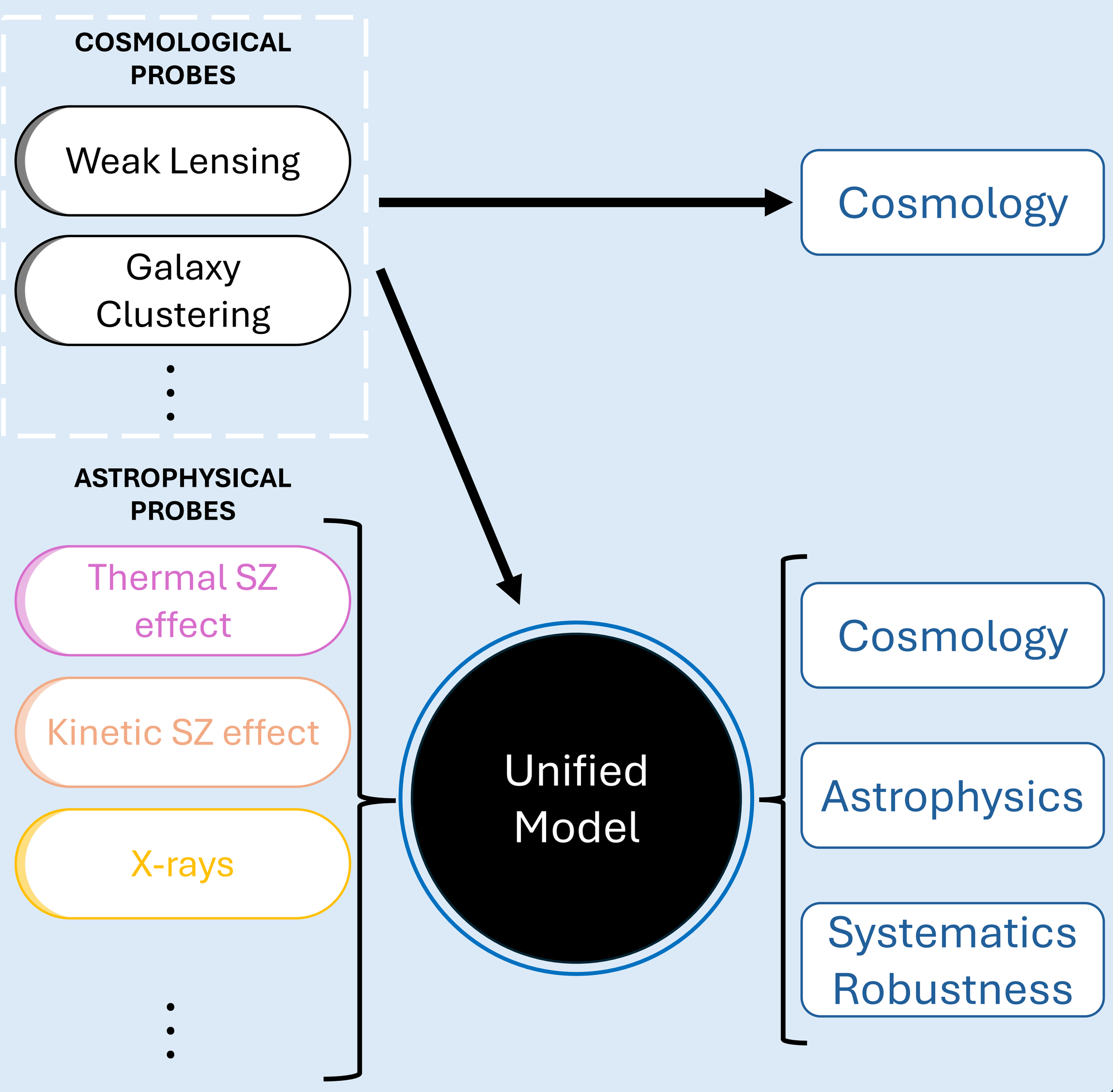
1. INTRODUCTION

Mapping out the large-scale structure is a core objective of Stage-IV experiments. The non-linear regime contains a wealth of information but accessing this information is limited by our understanding of the interplay between gravitational evolution and astrophysics.

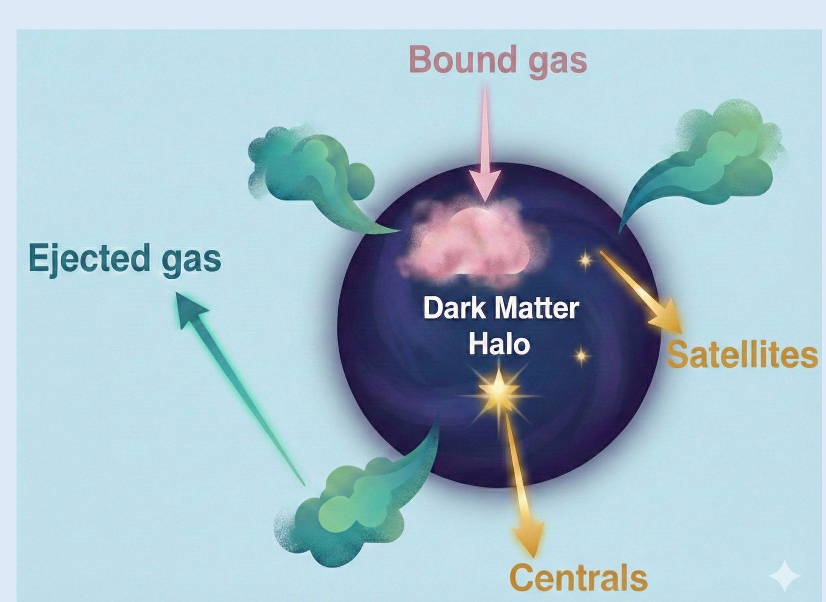


A unified model transforms baryonic systematic hurdles into a bridge between cosmology and galaxy evolution. This work assesses the readiness of current analytical tools for modeling cross-correlations across multi-wavelength datasets while preserving physical interpretability.



2. UNIFIED (HALO) MODELS OF COSMOLOGICAL FIELDS

- A halo model is an analytic framework that describes matter clustering by assuming all matter resides within discrete halos.
- The halo model relies on simulation-calibrated prescriptions for halo mass functions, linear bias, and concentration-mass relations.
- Recent extensions incorporate baryonic physics by adding gas and stellar components, introducing additional parameters to model their abundance and spatial distribution within the halo.



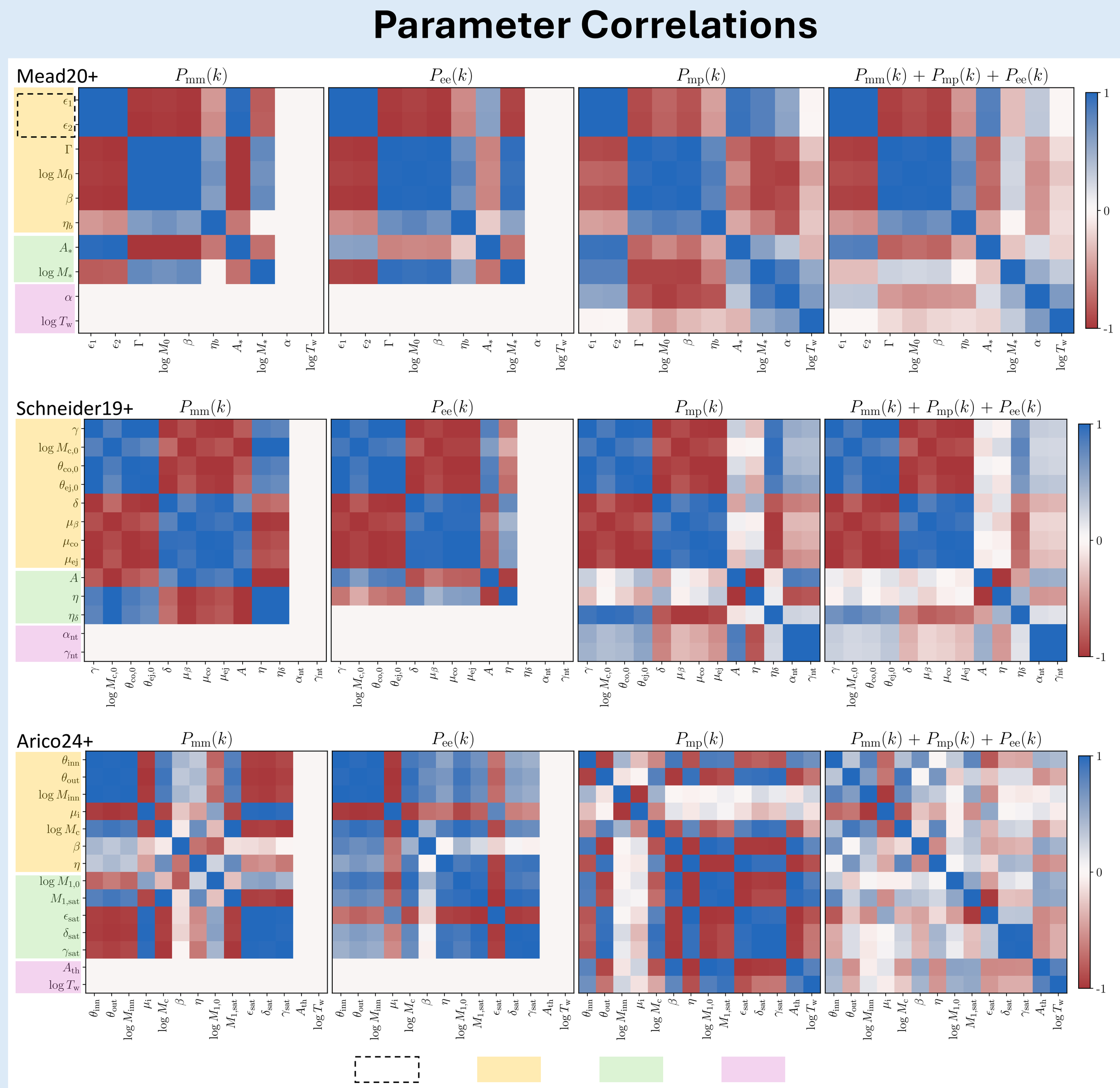
Models considered in this work

Mead20+ (Mead et al. 2020)
- Employs a simple prescription for the gas profile and non-thermal pressure.
- Requires *(unphysical)* smoothing parameter for modeling the transition regime
- The effect of feedback is accounted for by modifying halo concentration

Schneider19+ (Schneider et al. 2019)
- Halo profiles are untruncated; highly flexible gas density profile
- , naturally recovers power in the transition regime
- The gravitational back-reaction on the dark matter halo is included by assuming adiabatic relaxation

Arico24+ (Aricò & Angulo 2024)
- Highly flexible prescription for the gas density profile
- Sophisticated treatment of non-thermal pressure Requires *(unphysical)* smoothing parameter for modeling the transition regime
- Uses adiabatic relaxation to perturb halo profiles due to astrophysical feedback

Parameter Correlations



3. TESTING MODELS ON HYDRODYNAMICAL SIMULATIONS

Key questions
- Are analytical halo models sufficiently flexible to jointly and self-consistently describe cosmological and astrophysical observables?
- Do these models provide physical insight into the underlying baryonic properties?

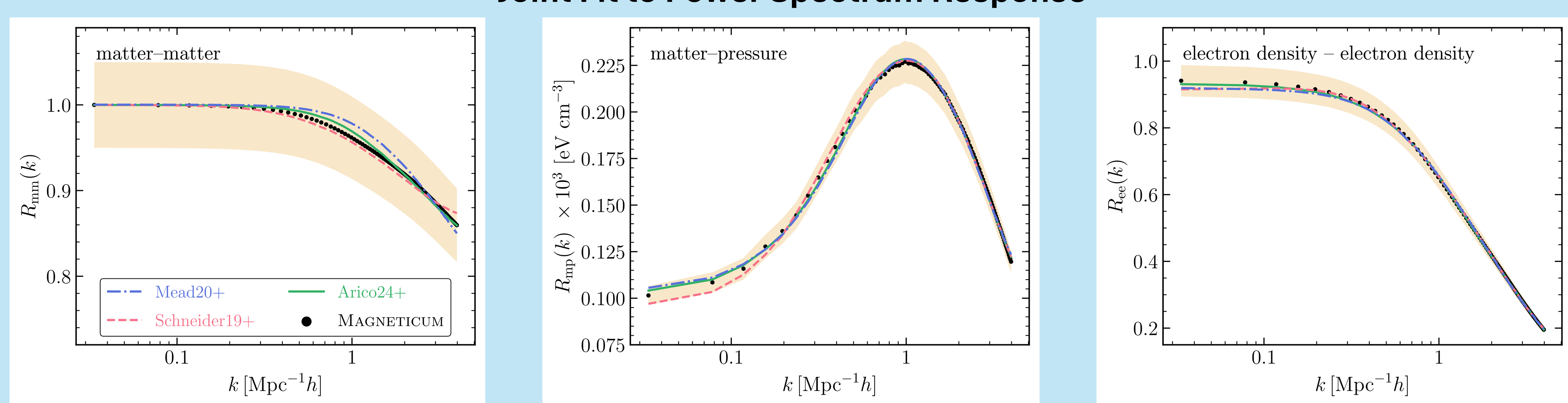
We evaluate model performance by jointly fitting the 3D power spectrum response underlying different tracers (cosmic shear, tSZ, kSZ) and testing the accuracy of the inferred halo properties.

The power spectrum response for fields A and B is defined as:

$$R_{AB}(k) = \frac{P_{AB}(k)}{P_{\mathrm{mm,DMO}(k)}}$$

We use the Box2b/hr *Magneticum* hydrodynamical simulation for measuring the power spectra and halo properties. *Magneticum* has not been used for calibrating any of the models and thus serves as an independent test case.

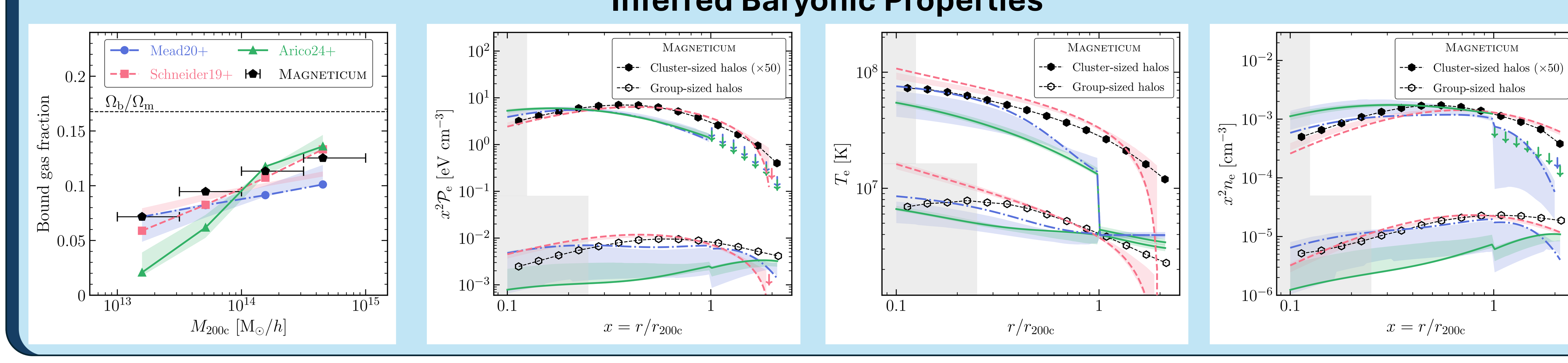
Joint Fit to Power Spectrum Response



All models achieve percent-level accuracy when jointly analyzing the total matter, pressure, and electron density fields

Models do not correctly infer the underlying gas abundance and thermodynamic profiles

Inferred Baryonic Properties



4. SUMMARY

- We benchmark three widely used analytical halo models, against an independent hydrodynamical simulation, in their ability to jointly describe the power spectra of cosmological fields (matter–matter, matter–pressure, and electron density-electron density).
- We find that percent-level fits to the power spectrum do not guarantee accurate recovery of baryonic properties: the error on the inferred thermodynamic profiles (gas pressure, density, etc) range from ~10% to ~50% across the three models.
- We identify key limitations and areas of refinement including 1-halo to 2-halo transition modeling, pressure-density connections, and non-thermal pressure treatment.

OUTLOOK

- Analytical approaches provide a computationally efficient framework for extracting cosmological and astrophysical constraints from multi-wavelength datasets (tSZ, kSZ, X-ray, weak lensing).
- However, analytical models lack internal consistency which can potentially lead to biased physical interpretation when applied to data.
- Analytical models can bridge cosmological and galaxy evolution studies, enabling cross-validation with targeted follow-up observations (X-ray, absorption lines) and leveraging galaxy-scale constraints on feedback.